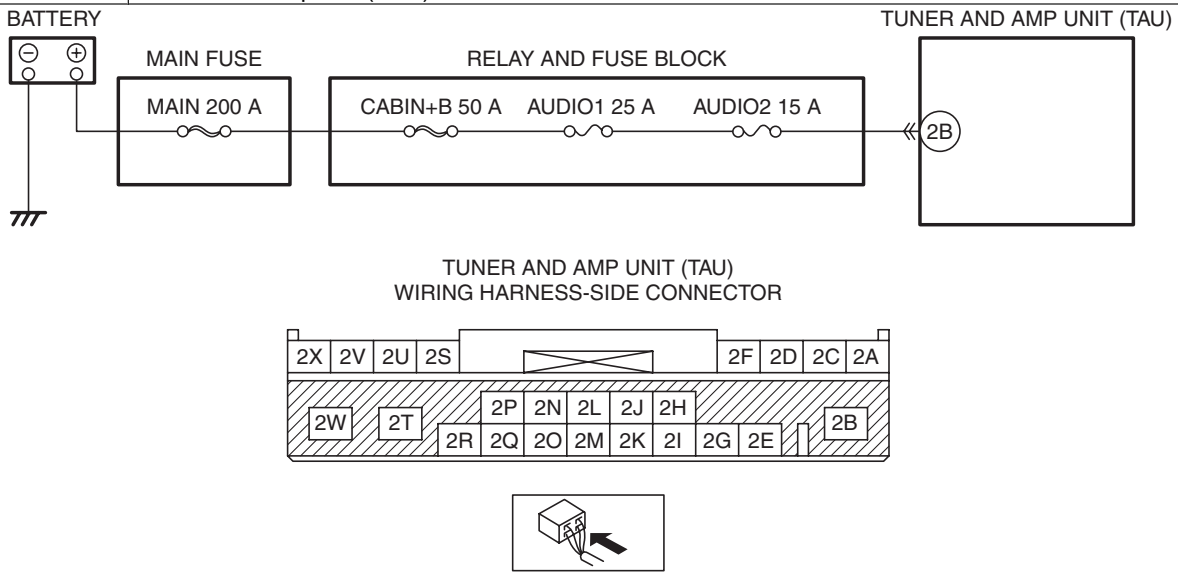
DTC B1238:16 [CONNECTIVITY MASTER UNIT]

id0902n7104900

Without i-stop System

Description	Tuner and amp unit (TAU) low power supply voltage input
Detection condition	• Connectivity master unit (CMU) detects power supply circuit voltage for tuner and amp unit (TAU) of **less than 10 V** for **10 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Tuner and amp unit (TAU) connector or terminal malfunction • Tuner and amp unit (TAU) power supply circuit malfunction — Short to ground in wiring harness between CABIN+B 50 A fuse and tuner and amp unit (TAU) terminal 2B — AUDIO2 15 A fuse malfunction — AUDIO1 25 A fuse malfunction — CABIN+B 50 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and tuner and amp unit (TAU) terminal 2B • Tuner and amp unit (TAU) malfunction



Diagnostic Procedure

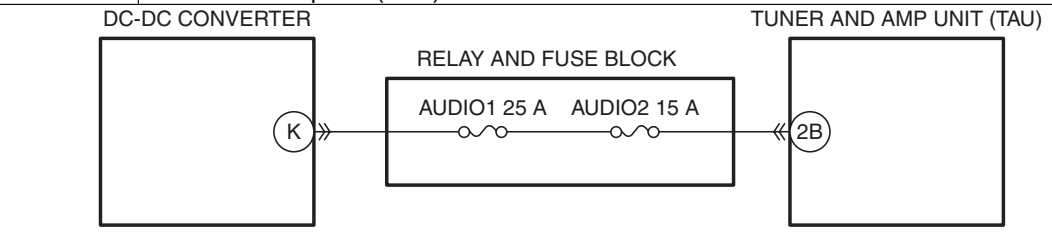
Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 6. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 6. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)

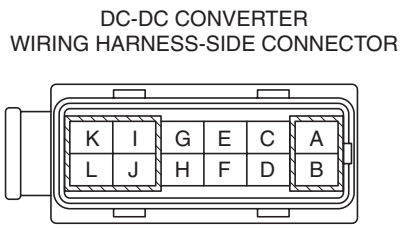
Step	Inspection		Action
4	**INSPECT TUNER AND AMP UNIT (TAU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the tuner and amp unit (TAU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
5	**VERIFY TUNER AND AMP UNIT (TAU) POWER SUPPLY VOLTAGE** • Verify that the tuner and amp unit (TAU) connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at tuner and amp unit (TAU) terminal 2B (wiring harness-side). • Is the voltage **10 V or more**?	Yes	Go to the next step.
		No	Inspect the AUDIO2 15 A fuse, AUDIO1 25 A fuse, and CABIN+B 50 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between CABIN+B 50 A fuse and tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between battery positive terminal and tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **17 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the tuner and amp unit (TAU). (See TUNER AND AMP UNIT (TAU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.

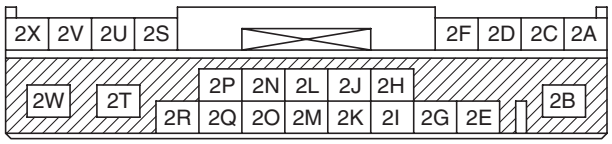
Step	Inspection		Action
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [CONNECTIVITY MASTER UNIT].)
		No	DTC troubleshooting completed.

With i-stop, without i-ELOOP

Description	Tuner and amp unit (TAU) low power supply voltage input
Detection condition	• Connectivity master unit (CMU) detects power supply circuit voltage for tuner and amp unit (TAU) of **less than 10 V** for **10 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Tuner and amp unit (TAU) connector or terminal malfunction • Tuner and amp unit (TAU) power supply circuit malfunction — Short to ground in wiring harness between AUDIO 1 25 A fuse and tuner and amp unit (TAU) terminal 2B — AUDIO2 15 A fuse malfunction — AUDIO1 25 A fuse malfunction — Open circuit in wiring harness between DC-DC converter terminal K and tuner and amp unit (TAU) terminal 2B • DC-DC converter connector or terminal malfunction • Tuner and amp unit (TAU) malfunction







Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 8. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

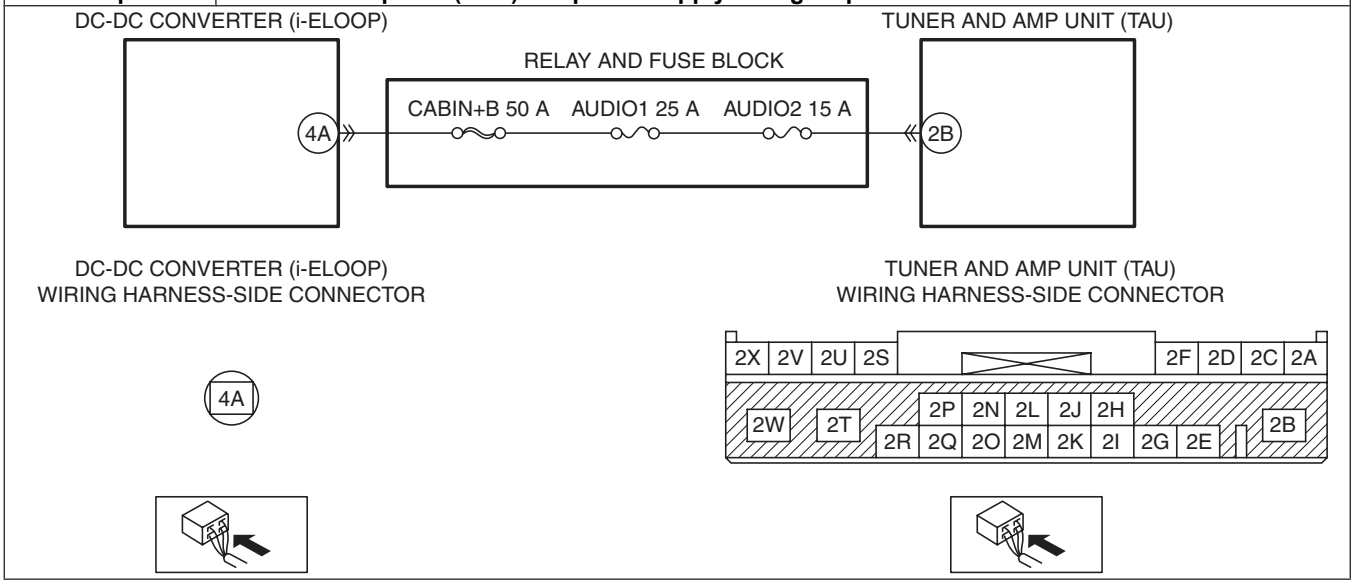
Step	Inspection		Action
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 8. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
4	**INSPECT TUNER AND AMP UNIT (TAU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the tuner and amp unit (TAU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.
5	**VERIFY TUNER AND AMP UNIT (TAU) POWER SUPPLY VOLTAGE** • Verify that the tuner and amp unit (TAU) connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at tuner and amp unit (TAU) terminal 2B (wiring harness-side). • Is the voltage **10 V or more**?	Yes	Go to the next step.
		No	Inspect the AUDIO2 15 A fuse and AUDIO1 25 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between AUDIO1 25 A fuse and tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between DC-DC converter terminal K and the tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.

Step	Inspection		Action
6	**INSPECT DC-DC CONVERTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the DC-DC converter connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.
7	**INSPECT DC-DC CONVERTER** • Inspect the DC-DC converter. (See DC-DC CONVERTER INSPECTION [WITHOUT i-ELOOP].) • Is the DC-DC converter normal?	Yes	Go to the next step.
		No	Replace the DC-DC converter, then go to Step 8. (See DC-DC CONVERTER REMOVAL/INSTALLATION [WITHOUT i-ELOOP].)
8	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **17 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the tuner and amp unit (TAU). (See TUNER AND AMP UNIT (TAU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
9	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [CONNECTIVITY MASTER UNIT].)
		No	DTC troubleshooting completed.

With i-ELOOP

Description	Tuner and amp unit (TAU) low power supply voltage input
Detection condition	• Connectivity master unit (CMU) detects power supply circuit voltage for tuner and amp unit (TAU) of **less than 10 V** for **10 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Battery malfunction • Generator malfunction • Tuner and amp unit (TAU) connector or terminal malfunction • Tuner and amp unit (TAU) power supply circuit malfunction — Short to ground in wiring harness between CABIN+B 50 A fuse and tuner and amp unit (TAU) terminal 2B — AUDIO2 15 A fuse malfunction — AUDIO1 25 A fuse malfunction — CABIN+B 50 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and tuner and amp unit (TAU) terminal 2B • Tuner and amp unit (TAU) malfunction

Description	Tuner and amp unit (TAU) low power supply voltage input



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
3	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 7. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5	**INSPECT TUNER AND AMP UNIT (TAU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the tuner and amp unit (TAU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.

Step	Inspection		Action
6	**VERIFY TUNER AND AMP UNIT (TAU) POWER SUPPLY VOLTAGE** • Verify that the tuner and amp unit (TAU) connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at tuner and amp unit (TAU) terminal 2B (wiring harness-side). • Is the voltage **10 V or more**?	Yes	Go to the next step.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the AUDIO2 15 A fuse, AUDIO1 25A fuse, and CABIN+B 50 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between CABIN+B 50 A fuse and tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and the tuner and amp unit (TAU) terminal 2B. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to the next step. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **17 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the tuner and amp unit (TAU). (See TUNER AND AMP UNIT (TAU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [CONNECTIVITY MASTER UNIT].)
		No	DTC troubleshooting completed.